Practical 14 - Solutions

December 5, 2024

Question 1: Describe the data *Use appropriate graphical tools and summaries to explore the variables in the data. Produce a table of baseline characteristics by treatment group. Summarise the main differences between DNase users and non-users.*

Before you start your analysis, it is good to familiarise yourself with the data. For example, you could look at histograms of the continuous variables and tabulate the binary/categorical variables. During this exploratory stage it's good to check for errors, sense check the values you have and investigate missing data.

In this case, we're interesting in investigating the effect of DNase on lung function, so we're comparing two groups: DNase users and non-users. It's useful here to present a table of patient characteristics by treatment group. This can be done using the following code:

```
#Baseline table
library(table1)
baseline_table<-table1(~ factor(Sex) + factor(Genotype) + Age + BMI + Baseline_FEV +
factor(Lung_Infection) + factor(HSS)| Dnase, data=CFdata)
baseline_table
```

	DNase non-users (n=9294)	DNase users (n=706)
Female	4520 (48.6%)	367 (52.0%)
High Genotype	2766 (29.8%)	200 (28.3%)
Age	23.0 (15.4)	34.1 (16.7)
BMI	17.9 (4.85)	18.7 (4.82)
FEV (at baseline)	53.4 (10.9)	45.3 (10.6)
Lung infection present	1912 (20.6%)	147 (20.8%)
HSS user	1818 (19.6%)	139 (19.7%)

Table 1: Table of baseline characteristics. Binary variables are summarised as N (%) continuous variables are summarised using mean (sd)

We have a sample of 10,000 individuals; 9294 are not taking DNase and 706 are taking DNase. The proportions of females, individuals with high genotype, individuals with lung infection and HSS users are similar between DNase users and non-users. On average, DNase users are older and have lower FEV at baseline.

Question 2: Univariable analysis *Use simple linear regression to estimate the effect of DNase on FEV. Give 95% confidence intervals for the estimated treatment effect and interpret the results.*

The following code can be used to run the univariable analysis:

```
summary(univariable_model<-lm(CFdata$FEV~CFdata$Dnase))
confint(univariable_model, parm=2, level=0.95)
```

The unadjusted estimated effect (and 95% CI) is 5.55 (3.97, 7.12). The expected FEV for people who take DNase is 5.55 higher than people who do not take DNase. Since 0 is not included in the confidence interval,

we have evidence of a positive effect of DNase on lung function. However, this is the unadjusted estimate and will be subject to confounding bias.

Question 3: Multivariable analysis *Use multivariable linear regression to estimate the effect of DNase on FEV, while adjusting for the listed confounders. Give 95% confidence intervals for the estimated treatment effects and interpret the results. Comparing the results between the multivariable analysis and the univariable analysis, what can we conclude?*

The following code can be used to run the multivariable analysis:

```
summary(multivariable_model<-lm(CFdata$FEV~CFdata$Dnase+CFdata$Age+CFdata$BMI+CFdata$Sex+
CFdata$Baseline_FEV+CFdata$Genotype+CFdata$Lung_Infection+
CFdata$HSS))
confint(multivariable_model, parm=2, level=0.95)
```

The adjusted estimated effect (and 95% CI) is 9.75 (8.18 11.33). After adjusting for the listed confounders, the expected FEV for people who take DNase is 9.75 higher than people who do not take DNase. Notice that the estimated effect is much higher in this analysis; the lower limit of the CI in the multivariable analysis is higher than the upper limit of the CI in the univariable analysis.

In the multivariable model, you may notice that age and baseline FEV are important predictors (very small p-values) and so they are strongly associated with the outcome. They are also strongly associated with the treatment (we noticed differences in the mean values of these variables between treatment groups in the baseline table). Therefore, baseline FEV and age are likely to be important confounders in this analysis. Notice that people who take DNase tend to be older, and older age is associated with a worse FEV. People who take DNase also tended to have worse FEV at baseline, and worse FEV at baseline is associated with worse FEV at follow-up. Therefore these confounders, when not accounted for, will have made it appear that DNase is not working as well as it should. This explains the smaller effect size observed in the univariable analysis.

Question 4: Checking assumptions Use appropriate graphical tools to check the validity of assumptions made in the multivariable linear regression model defined in Q3. Are there any influential observations?

The following code will obtain relevant diagnostic plots:

```
#Plot the residuals against the fitted values
plot(multivariable_model, which=1)

#Normal Q-Q plot of the standardised residuals
plot(multivariable_model, which=2)

#Plot of Cook's distances
plot(multivariable_model, which=4)
```

The plots are given at the end of this document.

Linearity and homoscedasticity can be assessed using the residual plot. Linearity seems to hold as the distribution of residuals is fairly equal above and below the red line. There are no obvious violations of the homoscedasticity assumption as the variance of residual appears to be approximately constant across the range of fitted values.

The QQ-plot can be used to assess normality. In this case, normality appears to hold as most of the points lie on the $Y = X$ line.

Finally, Cook's distance is a useful plot to detect outliers (these can usually be spotted in residual plots also). There do not appear to be any problematic outliers in this case.

Question 5: Further investigations *Use multivariable linear regression to investigate the following: (1) does the estimated treatment effect differ for males and females? (2) does the estimated treatment effect*

depend on FEV at baseline?

This question can be investigated using interaction terms.

(1) can be investigated by including an interaction term between DNase and sex:

```
summary(multivariable_model<-lm(CFdata$FEV~CFdata$Dnase*CFdata$Sex+CFdata$Age+CFdata$BMI+
CFdata$Baseline_FEV+CFdata$Genotype+CFdata$Lung_Infection+CFdata$HSS))
```

(2) can be investigated by including an interaction term between DNase at FEV at baseline:

```
summary(multivariable_model<-lm(FEV~Dnase*Baseline_FEV+Age+BMI+Genotype+Lung_Infection+HSS,
data=CFdata))
```

To assess whether there is evidence of an interaction, we can use the p-values associated with the regression parameter for the interaction term. A low p-value is strong evidence against the null hypothesis that the interaction term is equal to 0. In other words, a low p-value is evidence that there is an interaction.

In this case, there is little evidence that there is an interaction between sex and DNase. Therefore, we don't expect the treatment effect to be difference for males and females.

The p-value for the interaction term between baseline FEV and DNase is close to 0.05 and this could be considered as weak evidence that there is an an interaction. As the interaction term is positive, this suggests that people with high FEV at baseline benefit more from treatment than people with low FEV at baseline.

In this case, we could report the estimated effects for people with high FEV and low FEV separately. For example, the following code estimates the causal effect of DNase for people who had an FEV of 80 at baseline, and the causal effect of DNase for people who had an FEV of 20 at baseline.

```
#Predicted average outcome in the treated (high FEV)
treated_highFEV<-CFdata
treated_highFEV$Dnase<-"Yes"
treated_highFEV$Baseline_FEV<-80
predicted_treated_highFEV<-mean(predict(multivariable_model, newdata=treated_highFEV))
predicted_treated_highFEV

#Predicted average outcome in the untreated (high FEV)
untreated_highFEV<-CFdata
untreated_highFEV$Dnase<-"No"
untreated_highFEV$Baseline_FEV<-80
predicted_untreated_highFEV<-mean(predict(multivariable_model, newdata=untreated_highFEV))
predicted_untreated_highFEV

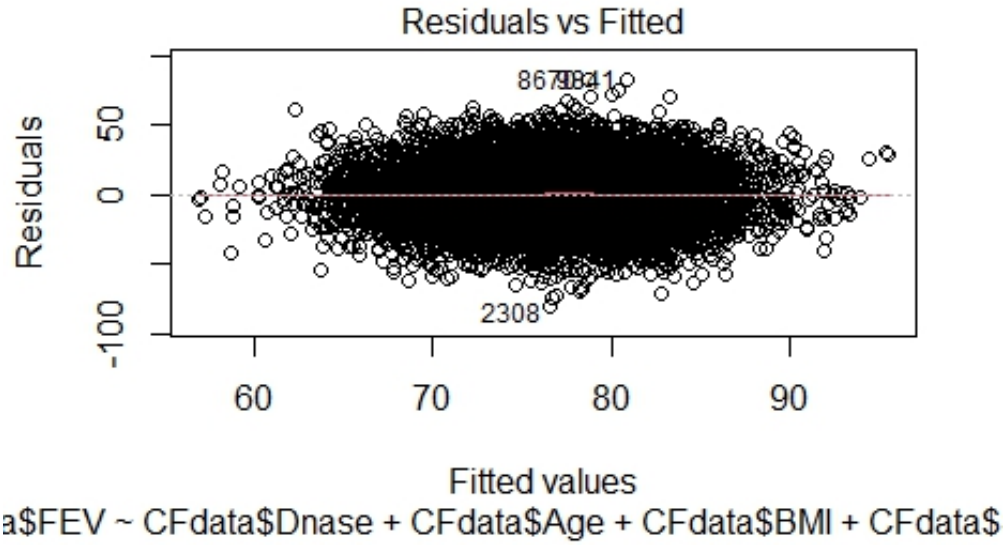
#Predicted average outcome in the treated (low FEV)
treated_lowFEV<-CFdata
treated_lowFEV$Dnase<-"Yes"
treated_lowFEV$Baseline_FEV<-20
predicted_treated_lowFEV<-mean(predict(multivariable_model, newdata=treated_lowFEV))
predicted_treated_lowFEV

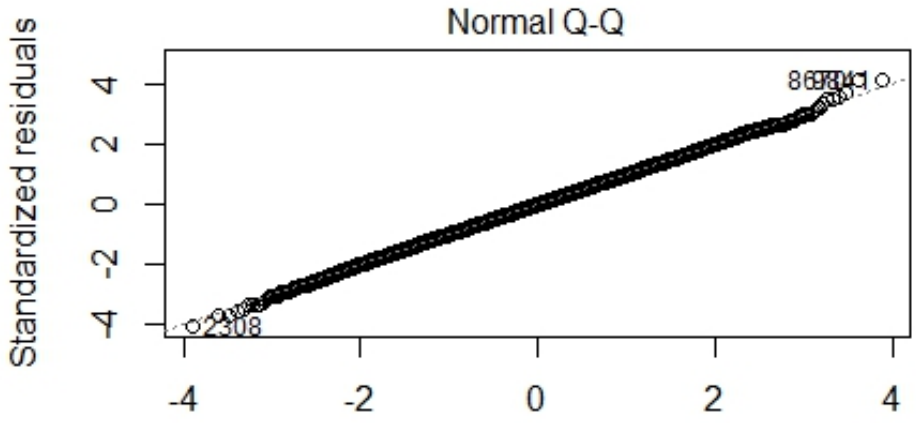
#Predicted average outcome in the untreated (low FEV)
untreated_lowFEV<-CFdata
untreated_lowFEV$Dnase<-"No"
untreated_lowFEV$Baseline_FEV<-20
predicted_untreated_lowFEV<-mean(predict(multivariable_model, newdata=untreated_lowFEV))
predicted_untreated_lowFEV

#Causal effect estimates
```

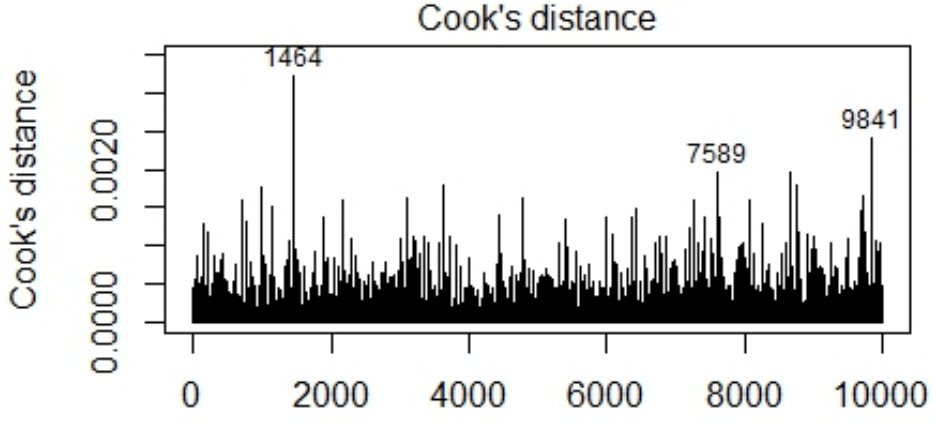
```
CE_low<-predicted_treated_lowFEV-predicted_untreated_lowFEV
CE_high<-predicted_treated_highFEV-predicted_untreated_highFEV
```

The estimated effect of DNase on lung function for people with low FEV is 6.12 and the estimated effect of DNase on lung function for people with high FEV is 14.49.





a$FEV ~ CFdata$Dnase + CFdata$Age + CFdata$BMI + CFdata$



a$FEV ~ CFdata$Dnase + CFdata$Age + CFdata$BMI + CFdata$